

AMIR JOUDAKI

Foundations of AI, AI for biomedicine, NLP/LLMs

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🔗 <https://ajoudaki.github.io/>

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EXPERIENCE

Postdoctoral researcher

ETH Zurich

📅 Nov 2024 – now

📍 Zürich, Switzerland

I work towards a comprehensive understanding of how deep nonlinear models learn. This includes understanding how modules interact and affect forward passes, backward gradients, and the neural kernel evolution in depth.

EDUCATION

Direct Ph.D. in Artificial Intelligence

ETH Zurich

📅 Feb 2017 – now

📍 Zurich, Switzerland

Direct PhD is a highly selective program (<5% of all PhD admissions), with 2 years of MSc courses followed by 4 years of PhD research.

PhD Thesis: *On a mathematical understanding of deep neural networks AI for biomedicine.*

Authored 8 papers (6 first authored) in top-tier venues in AI/ML and computational genomics. Supervised over 10 MSc thesis/semester projects, organized teaching.

Ph.D. supervisors: Gunnar Ratsch & Francis Bach.

MSc Thesis: *Scalable algorithms for biological sequence analysis*

MSc GPA 5.68/6 (top 10%)

Algorithms Lab (5.75/6)

M.Sc. in Cognitive Neuroscience

International School for Advanced Studies (SISSA)

📅 Feb 2014 – Jan 2017

📍 Trieste, Italy

Thesis: *Modeling activity of electrophysiological recordings in vivo in rats*

B.Sc. in Computer Engineering

Sharif University of Technology

📅 Sept 2008 – Sept 2011

📍 Tehran, Iran

PROGRAMMING SKILLS

Python

PyTorch

JAX

Scikit-Learn

Numpy

C/C++

OpenMP

MPI

CUDA

Numba

Docker

Spark

SQL

Bash

Git

Java

LaTeX

AWARDS



Direct Doctorate Fellowship

Among two out of more than one hundred students selected for direct PhD program at ETH computer science department, 2017



Ranked 42nd

Among more than 50,000 participants in National Higher Education Entrance Exam, 2011.



Ranked 369th

Among more than 400,000 participants in National University Entrance Exam, 2007.

LANGUAGES

English: C2/Native

Persian: Native

German: A1.2

PUBLICATIONS

Mathematical foundations of AI

- [Amir Joudaki](#), Thomas Hofmann, “Emergence of globally attracting fixed points in deep neural networks with nonlinear activations”, AISTATS (poster).
- Alex Meterez*, [Amir Joudaki](#)*, Francesco Orabona, Alex Immer, Gunnar Ratsch, Hadi Daneshmand, “Batch normalization without gradient explosion: Towards training without depth limits”, accepted to proceedings of ICLR 2024 (poster). (*: equally contributed).
- [Amir Joudaki](#), Hadi Daneshmand Francis Bach, “On the impact of activation and normalization in obtaining isometric embeddings at initialization”, NeurIPS 2023 (poster).
- [Amir Joudaki](#), Hadi Daneshmand Francis Bach, “On Bridging the Gap between Mean Field and Finite Width in Deep Random Neural Networks with Batch Normalization”, ICML 2023 (poster).
- Hadi Daneshmand, [Amir Joudaki](#), Francis Bach, “Batch Normalization Orthogonalizes Representations in Deep Random Networks”, **spotlighted NeurIPS 2021 (top 3% of submissions)**.
- Alexandre Bense, [Amir Joudaki](#), Tim G. J. Rudner, Vincent Fortuin, “PCA Subspaces Are Not Always Optimal for Bayesian Learning”, NeurIPS 2021 workshop (DistShift).

Genomics Sequence Analysis

- Flavia Pedrocchi, Stefan Stark, Gunnar Ratsch, [Amir Joudaki](#), “Identifying Biological Priors and Structure in Single-Cell Foundation Models”, ICML workshop 2024 (Efficient and Accessible Foundation Models for Biological Discovery).
- Kacper Kapusniak, Manuel Burger, Gunnar Ratsch, [Amir Joudaki](#), “Learning Genomic Sequence Representations using Graph Neural Networks over De Bruijn Graphs”, NeurIPS workshop 2023 (Graph Learning Frontiers).
- [Amir Joudaki](#)*, Alexandru Meterez*, Harun Mustafa, Ragnar Groot Koerkamp, Andre Kahles, Gunnar Raetsch, “Aligning distant sequences to graphs using long seed sketches”, Journal of Genome Research 2023 (Cold Spring Harbor Laboratory press). (*: equally contributed).
- [Amir Joudaki](#), Gunnar Ratsch, Andre Kahles, “Fast Alignment-Free Similarity Estimation By Tensor Sketching”, proceedings of RECOMB 2021.
- Mikhail Karasikov, Harun Mustafa, [Amir Joudaki](#), Sara Javadzadeh-No, Gunnar Ratsch, André Kahles, “Sparse Binary Relation Representations for Genome Graph”, Journal of computational biology: a journal of computational molecular cell biology 27.4 (2020): 626-639.

Dimensionality Reduction

- Amir Najafi, [Amir Joudaki](#), “Nonlinear Dimensionality Reduction via Path-Based Isometric Mapping”, IEEE transactions on pattern analysis and machine intelligence, 38(7), 1452-1464.

Functional Brain Networks Analysis

- [Amir Joudaki](#)*, Elham Barzegaran*, Mahdi Jalili*, Andrea O. Rossetti, Richard S. Frackowiak, and Maria G. Knyazeva, “Properties of functional brain networks affect frequency of psychogenic non-epileptic seizures”, Frontiers in Human Neuroscience, 6:335. (*: equally contributed).
- [Amir Joudaki](#), Niloufar Salehi, Mahdi Jalili, Maria Knyazeva, “EEG-based functional brain networks: does the network size matter?”, PLoS ONE 7(4): e35673.

REFERENCES

Thomas Hofmann, Professor of computer science at ETH, Data Analytics lab, Zurich, Switzerland

Francis Bach, Professor of Computer Science department at INRIA in the Ecole Normale Supérieure, in Paris, France.

Gunnar Ratsch, Professor of computer science at ETH, Biomedical Informatics lab, Zurich, Switzerland

Francesco Orabona, Associate Professor at Boston University, Boston, USA.

Ragnar Martens, Team Lead Product at SkillLab, Amsterdam, Netherlands.